

Inductive Biases

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$$f(x) = 2x - 1 \qquad \text{(odd numbers)}$$

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The Need for Biases in Learning Generalizations

Tom M. Mitchell (1980)

*Learning involves the ability to generalize from past experience in order to deal with new situations that are "related to" this experience. **The inductive leap needed to deal with new situations seems to be possible only under certain biases** for choosing one generalization of the situation over another.*

No free lunch theorems for optimization

Wolpert (1996); Wolpert & Macready (1997)

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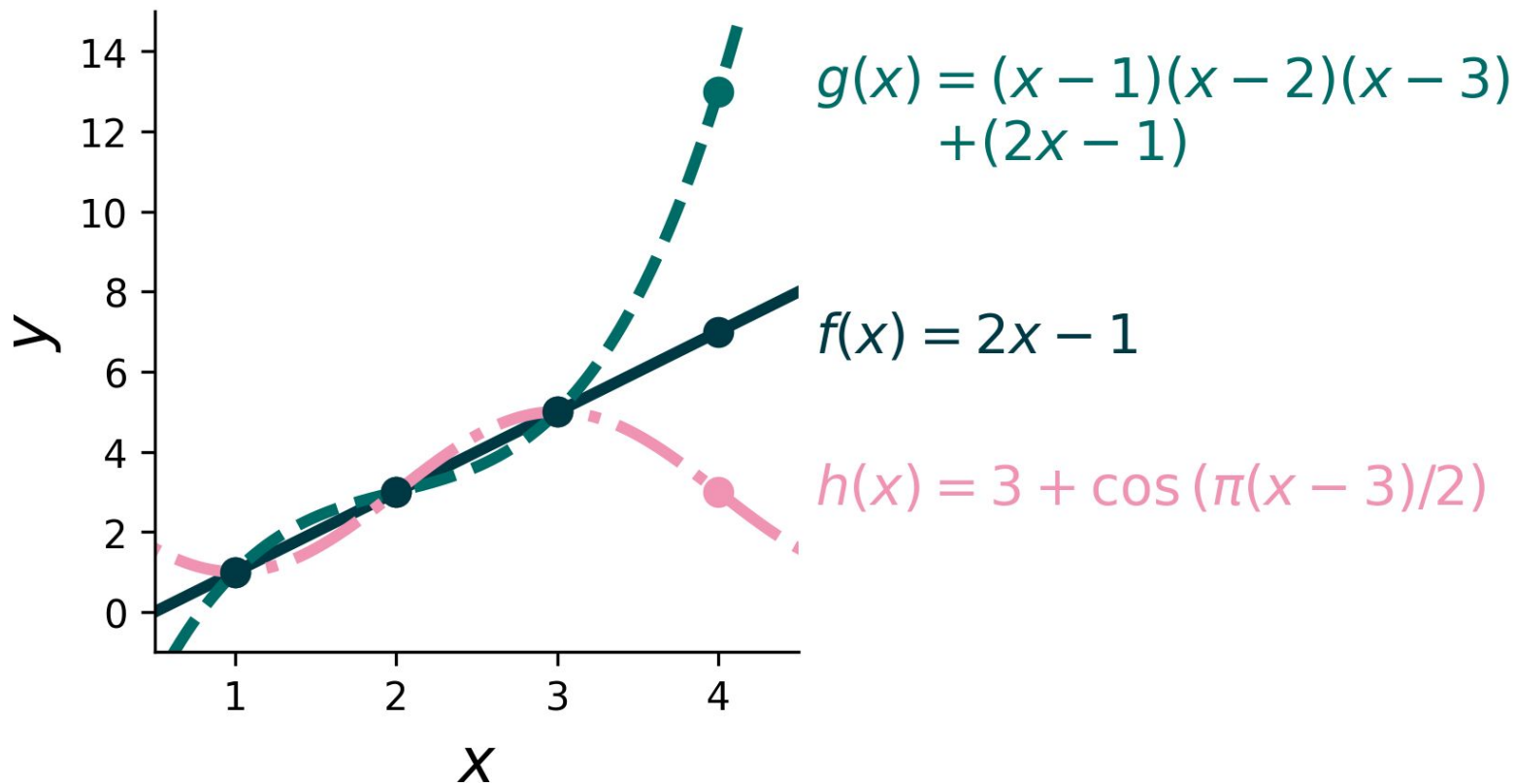
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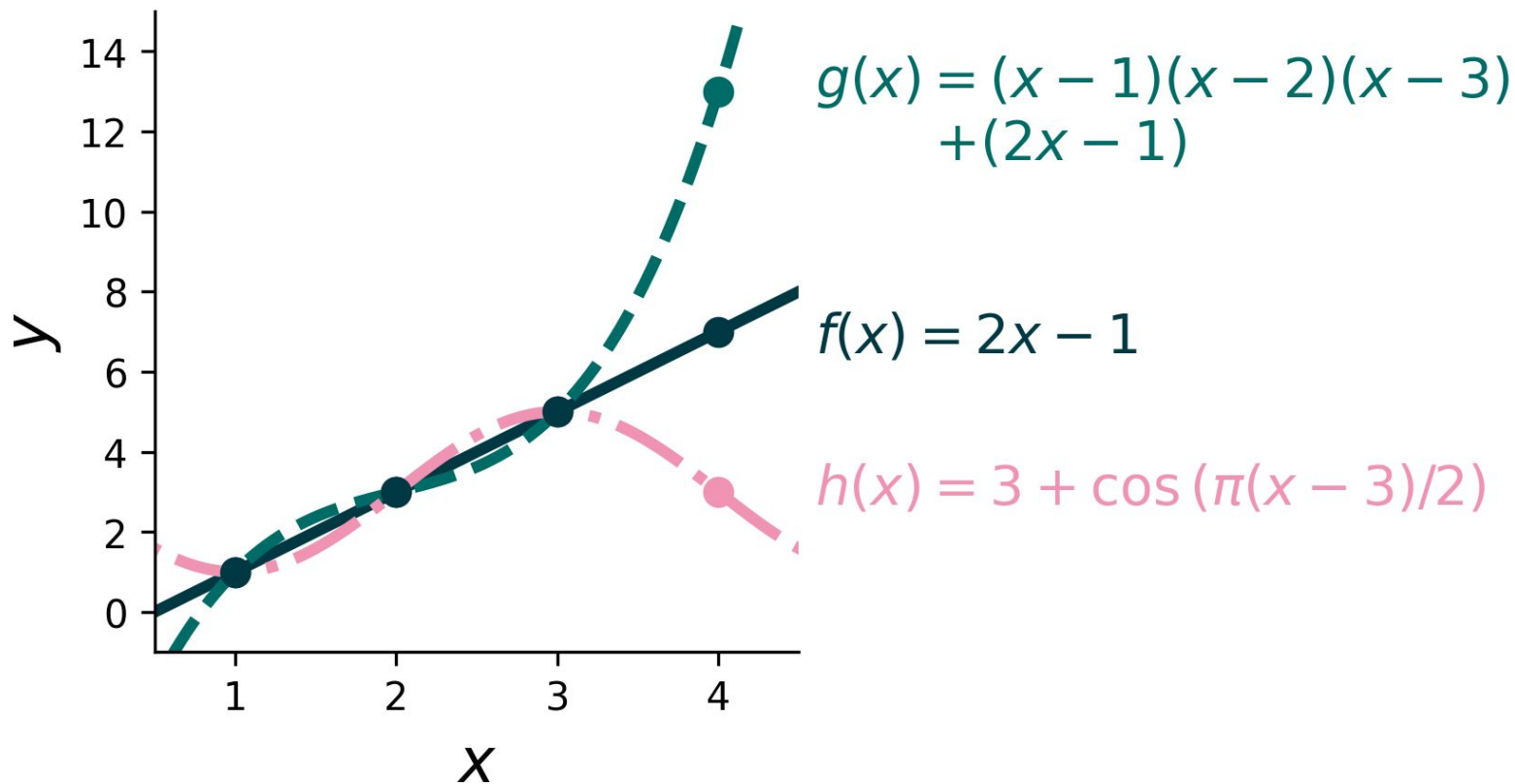
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But if we are *biased* towards solving certain kinds of problems, then certain algorithms are indeed better!

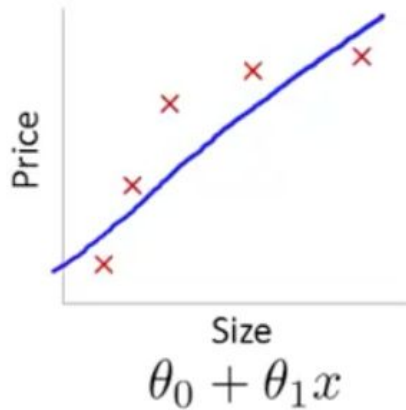
Which bias do you want? Simple? Cubic? Periodic?



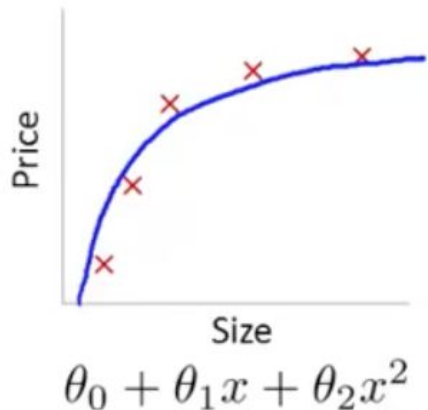
Inductive reasoning requires biases (assumptions)!



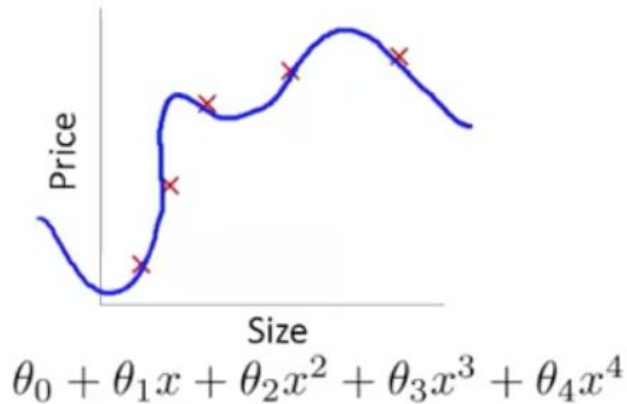
The classical bias–variance tradeoff



High bias
(underfit)
 $d=1$

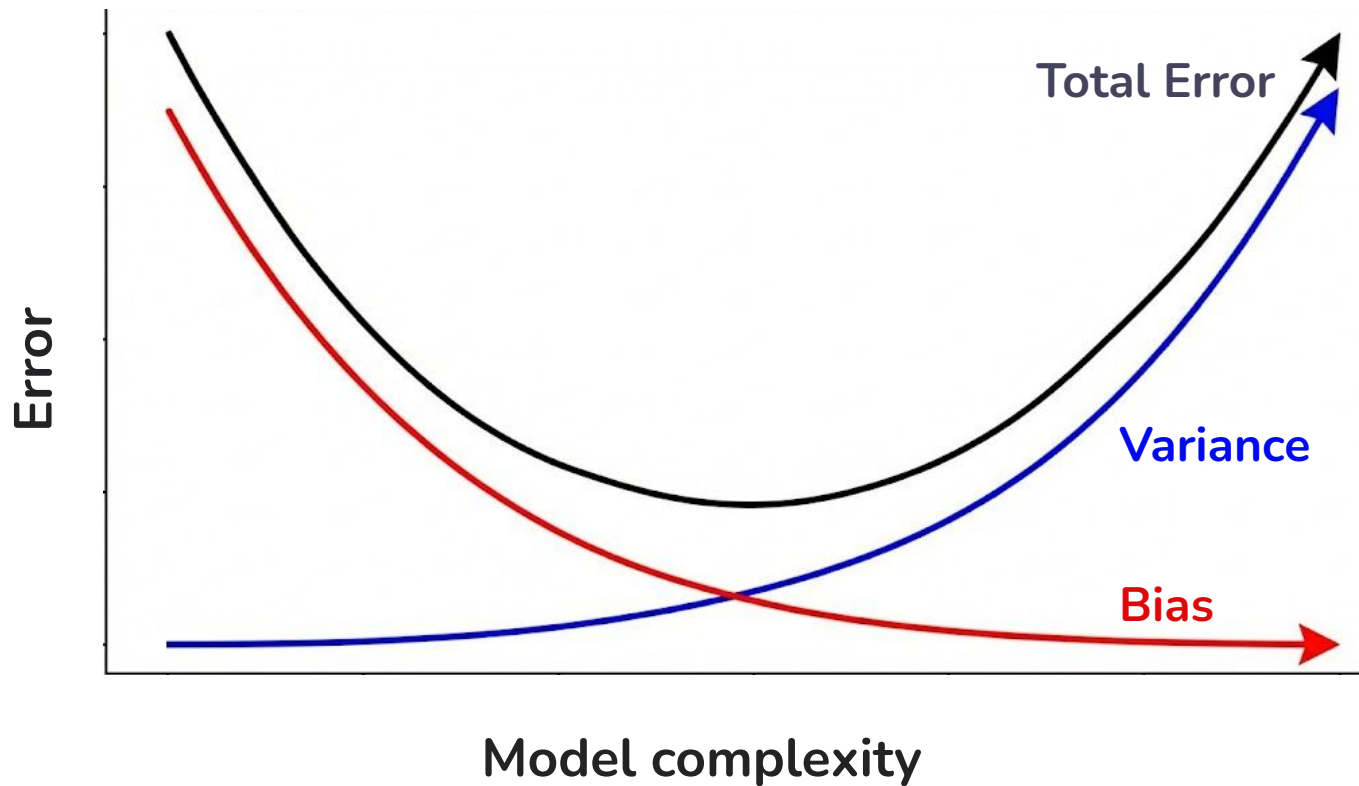


“Just right”
 $d=2$

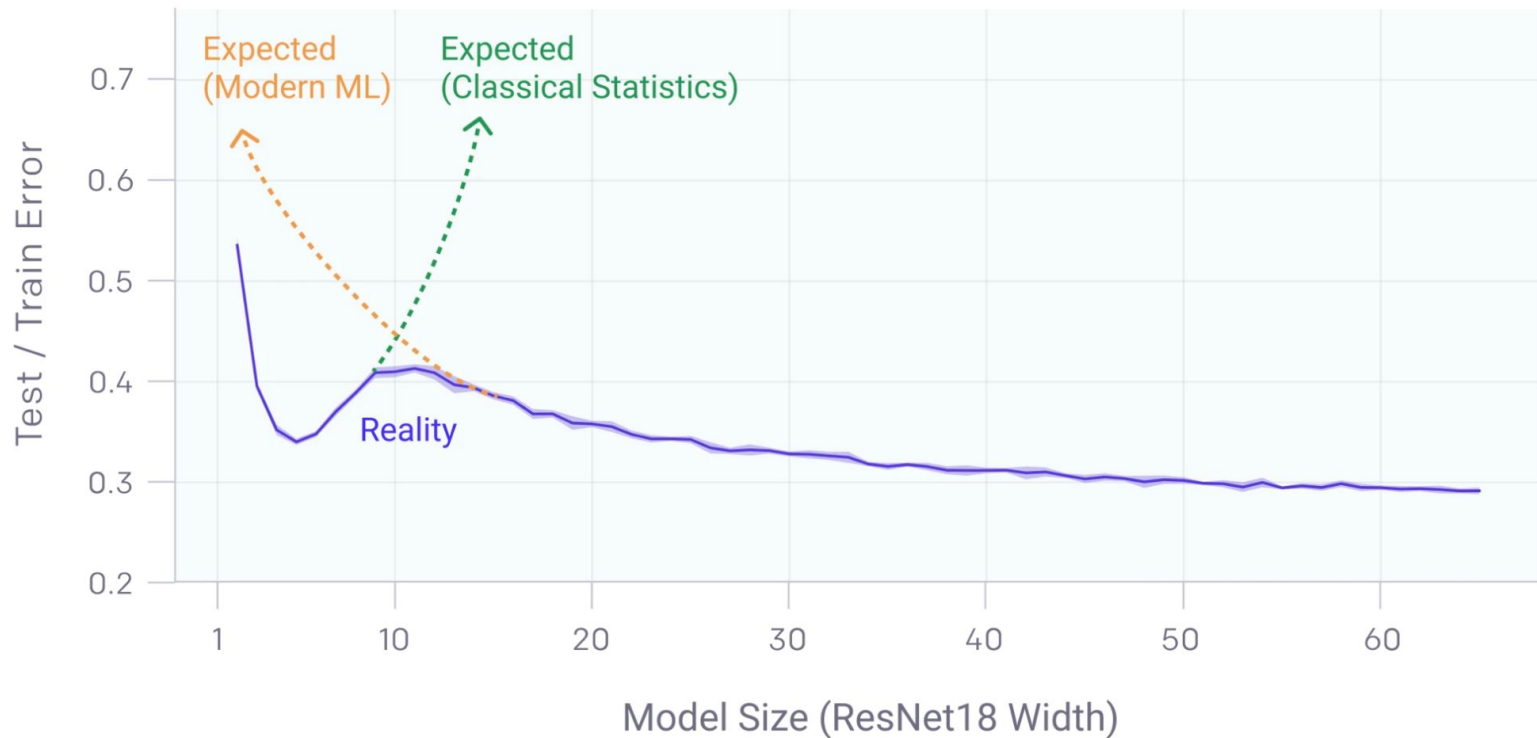


High variance
(overfit)
 $d=4$

The classical bias–variance tradeoff



Deep double descent:



Inductive biases in the deep learning era

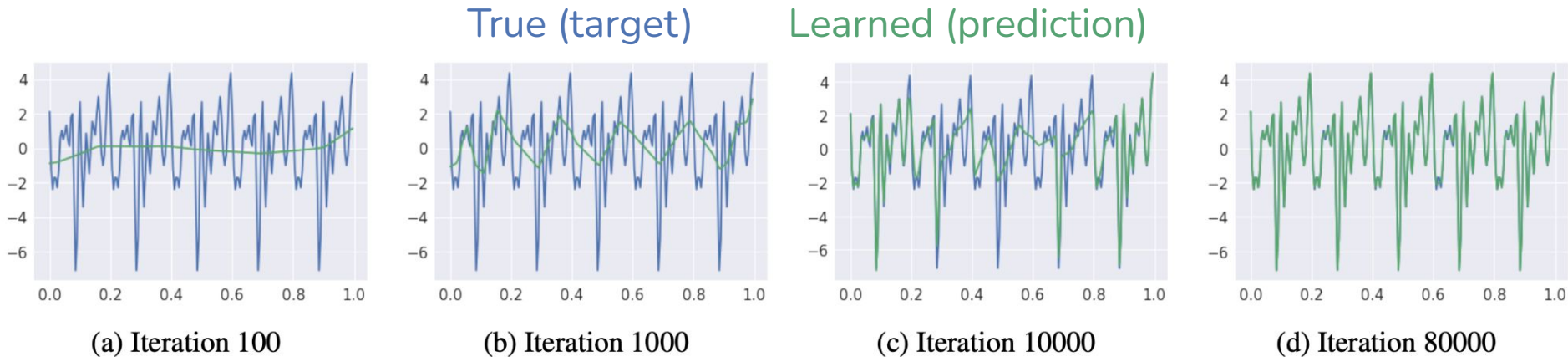
Deep neural networks easily fit random labels. (Zhang et al. 2016)

Deep ReLU networks are biased toward low frequency functions (Jacot et al. 2018) — they first learn large-scale, then small-scale features!

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One consequence of this: you will eventually fit the highest-frequency modes, ***including noise***, unless you:

- stop optimization at just the right time ("early stopping")
- alter the inductive biases to avoid fitting noise ("regularization")

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Spectral bias. Deep neural networks tend to learn low-frequency (smooth) features faster than high-frequency (noisy) ones.

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Locality, sequentiality, and periodicity. Light curves depend on recent time steps, and may exhibit repetition over fixed periods.

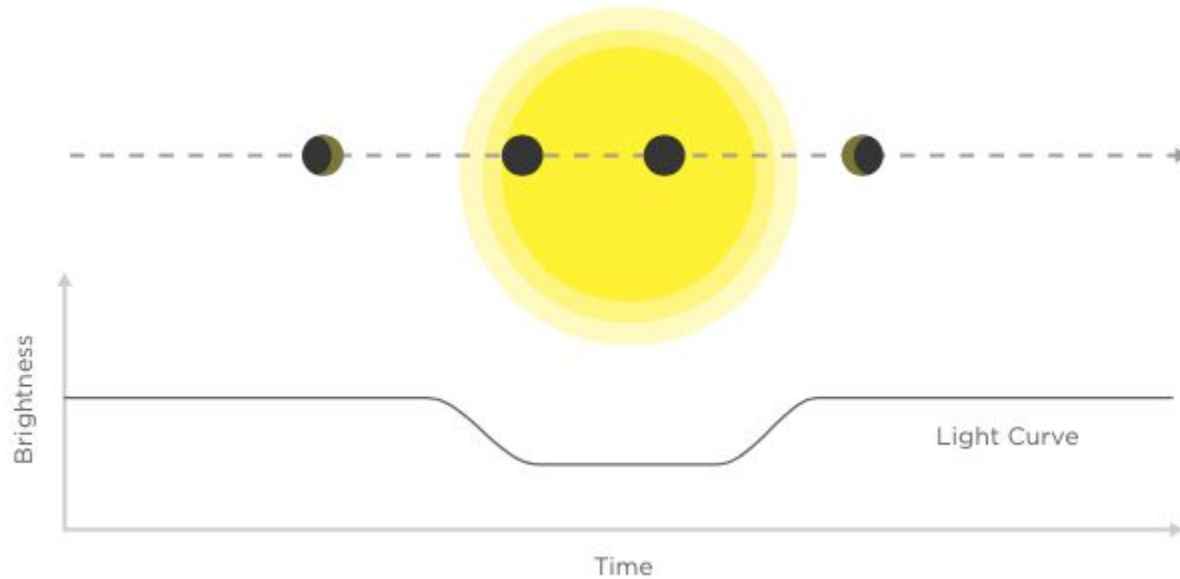
Conservation laws. Physical quantities can be conserved, such as mass or momentum or energy or phase-space volume.

Inductive reasoning requires biases (assumptions)

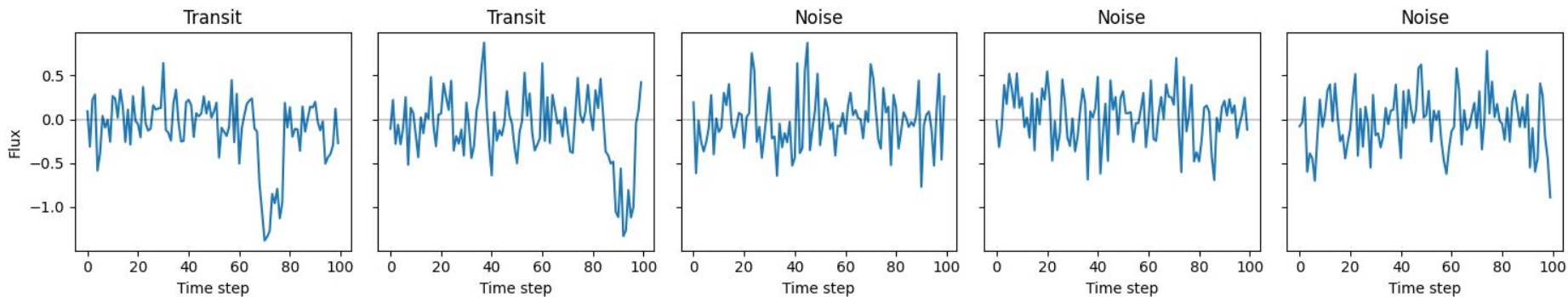
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"What kind of problem are you trying to solve?"

Example: classifying transits in light curves



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*Should the model care about **when** the transit occurs?*

*Are **fluxes independent of time**, or do we care about **correlations**?*

*Is it important to model **one time step after another**?*

*Do we care about **long-range (non-local) correlations**?*

Common NN architectures and their inductive biases

Multilayer Perceptron (MLP). Treats inputs as independent features, $\mathbf{x}$, and learns some model $y = f_{\theta}(\mathbf{x})$.

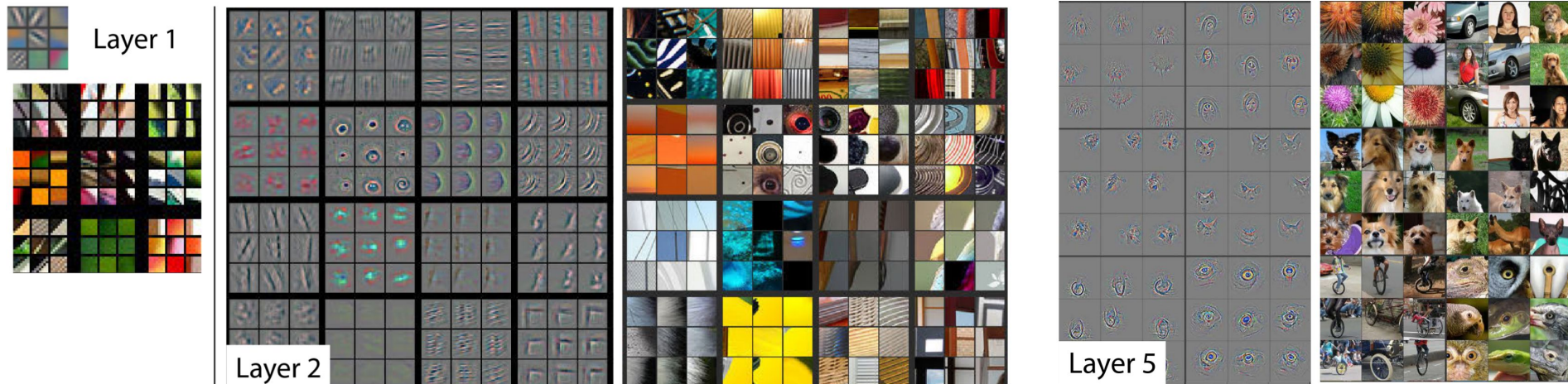
Weak inductive bias; treats all inputs independently.

r mag	NUV – r	axis_ratio	R ₅₀	concen.	inclination	gas mass frac.
15.437700	2.830000	0.612988	5.040000	2.550000	53.700000	-0.687000
15.010400	5.800000	0.762775	4.830000	3.050000	41.300000	-1.621000
15.183700	5.610000	0.620971	3.540000	3.250000	53.100000	-1.801000
13.964900	5.480000	0.286419	5.240000	2.650000	77.900000	-1.809000
14.459100	3.690000	0.445285	5.660000	2.330000	66.000000	-0.997000

Common NN architectures and their inductive biases

Convolutional neural network (CNN). Compositional matched-filter operations with empirically learned features.

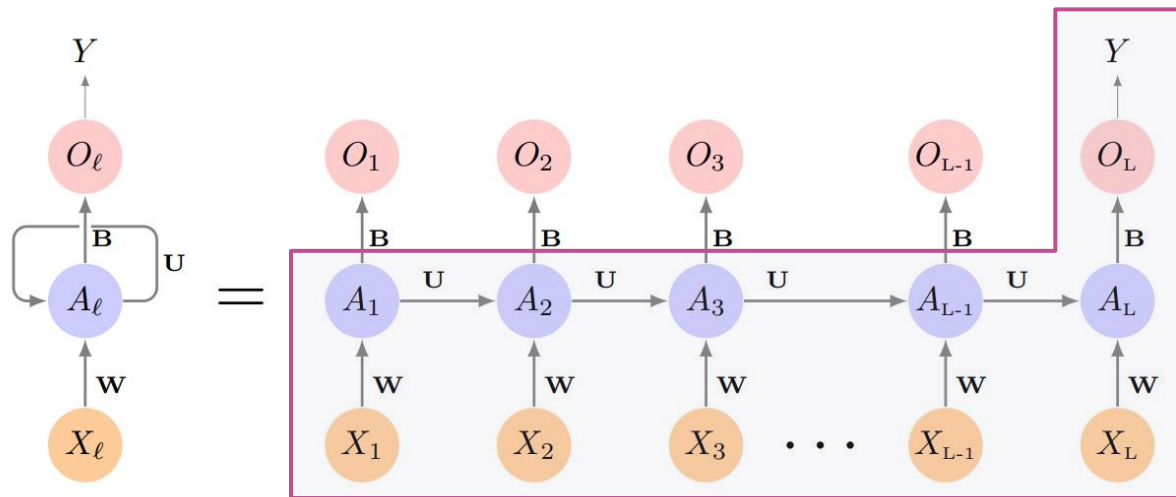
Inductive bias: local spatial structure, translation invariance of filters.



Common NN architectures and their inductive biases

Recurrent neural network (RNN). Models sequential structure where time-ordering is important, maintains a hidden state to model the past.

Inductive bias: sequentiality (but bad for long sequences), locality.



Common NN architectures and their inductive biases

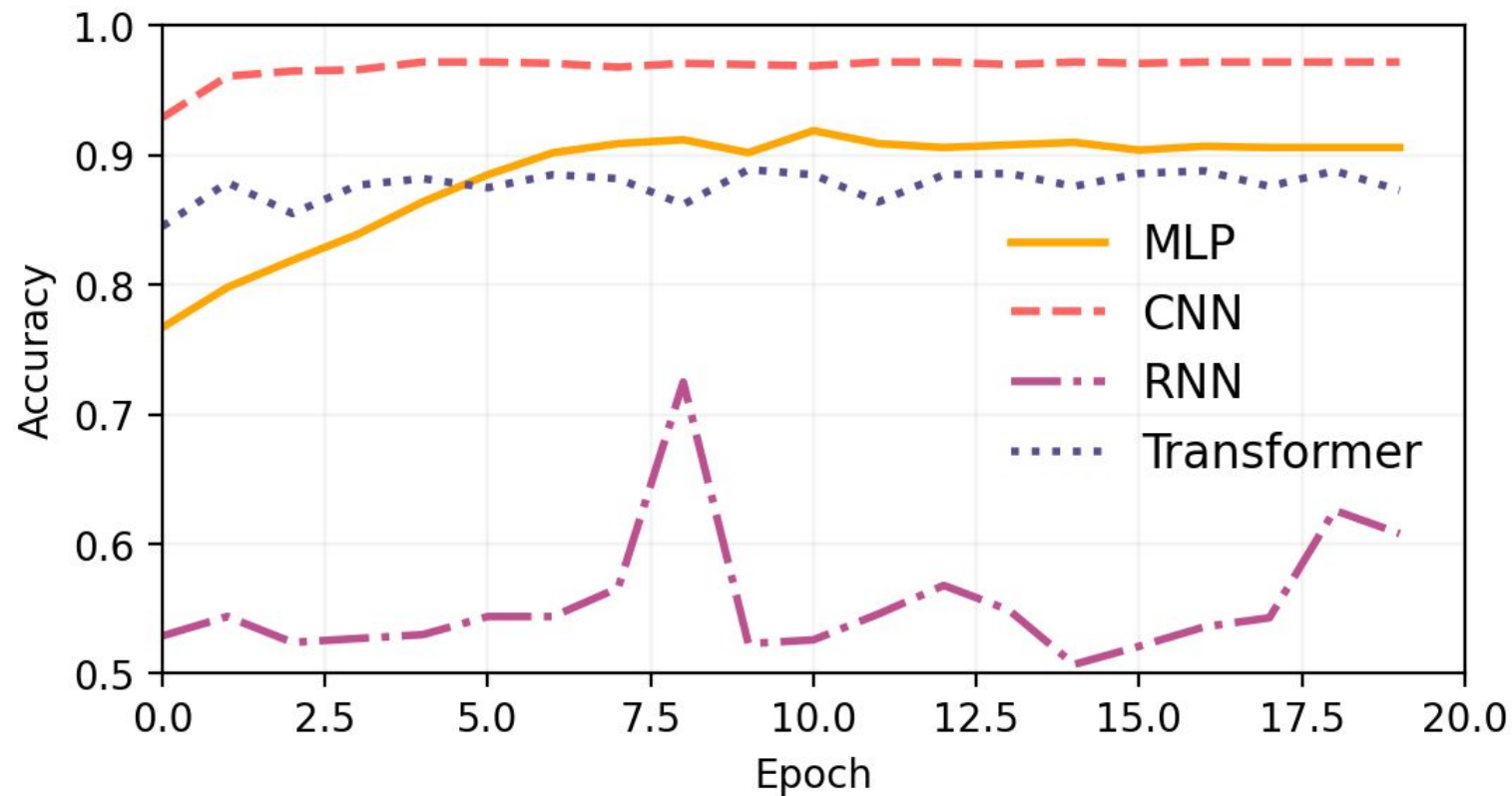
Transformers. Learns global interactions via attention without any inherent bias towards locality. Must manually add position information.

Inductive bias: global correlations, position-agnostic — data hungry!



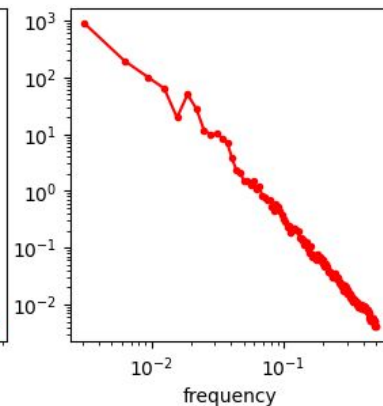
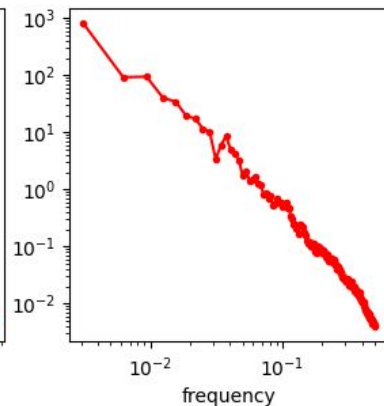
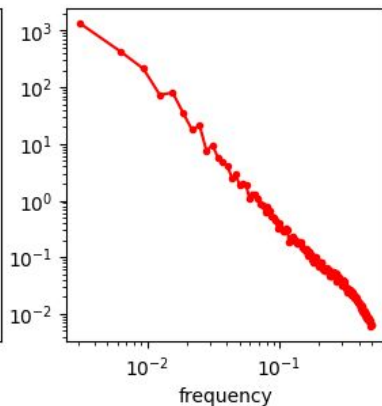
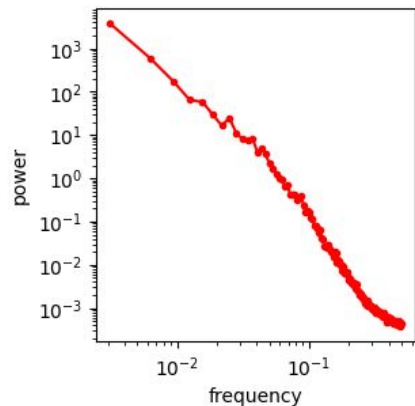
[Go to Colab Jupyter Notebook](#)

Inductive biases: example results

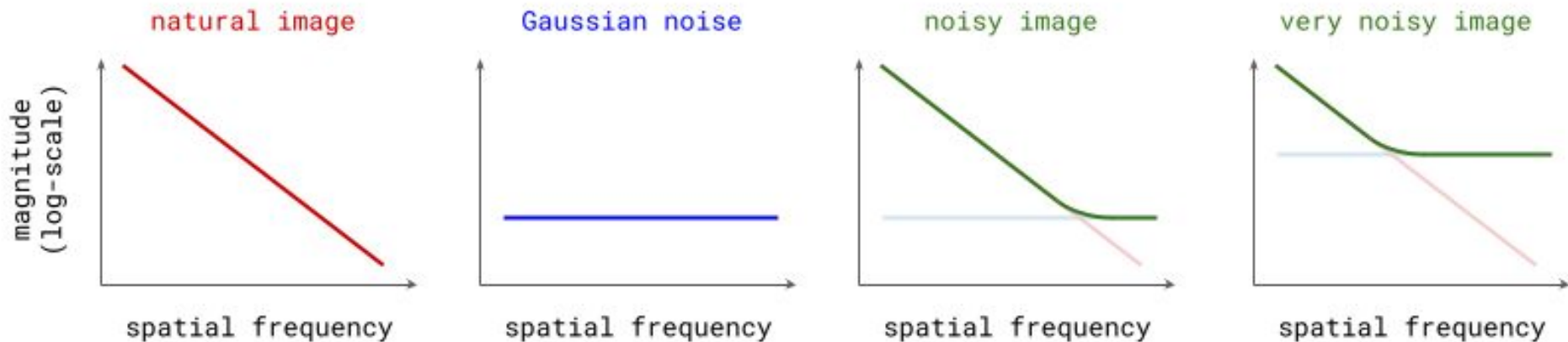


Aside: inductive biases for generative models

Natural images have power-law spectral densities



Adding Gaussian noise erases small-scale info



Diffusion models enabled by scale separation!

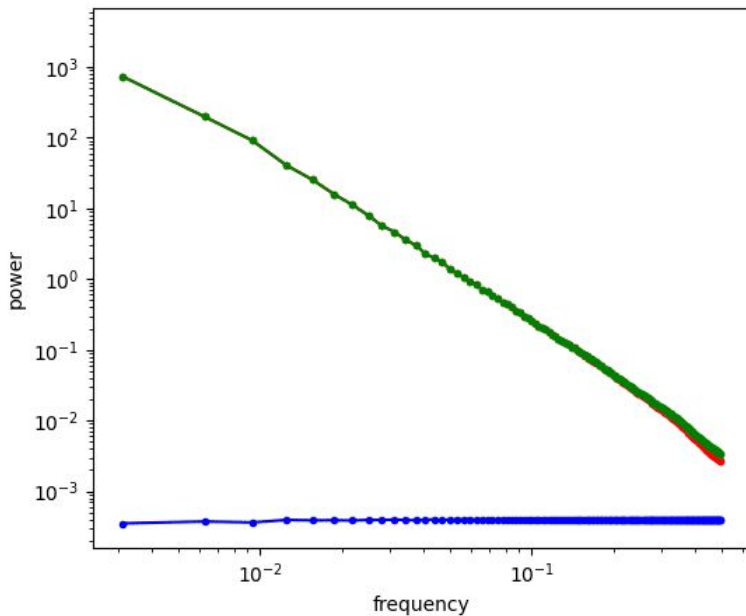
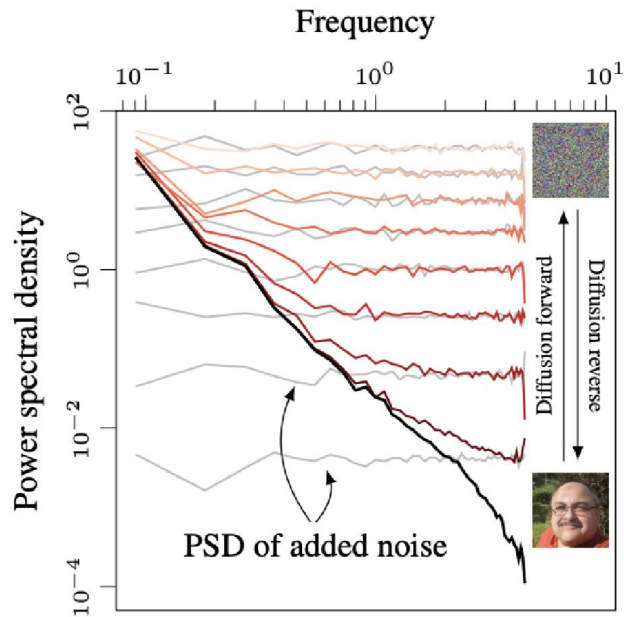


Figure 4: The $1/f^\alpha$ power spectral density in natural images induces an implicit coarse-to-fine inductive bias in diffusion models.